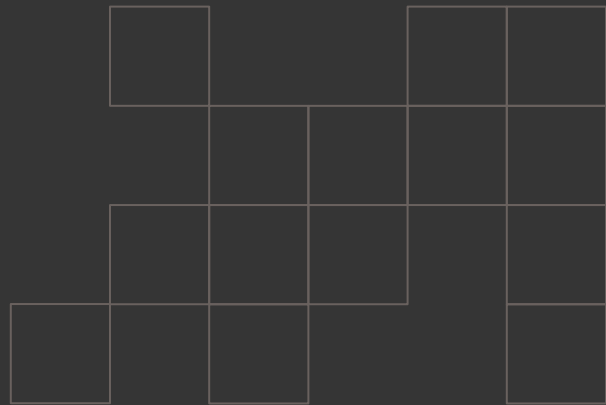


Modeling Fairness: How Reseeding MLB playoffs changes the game

Sofia Caceres



Background:

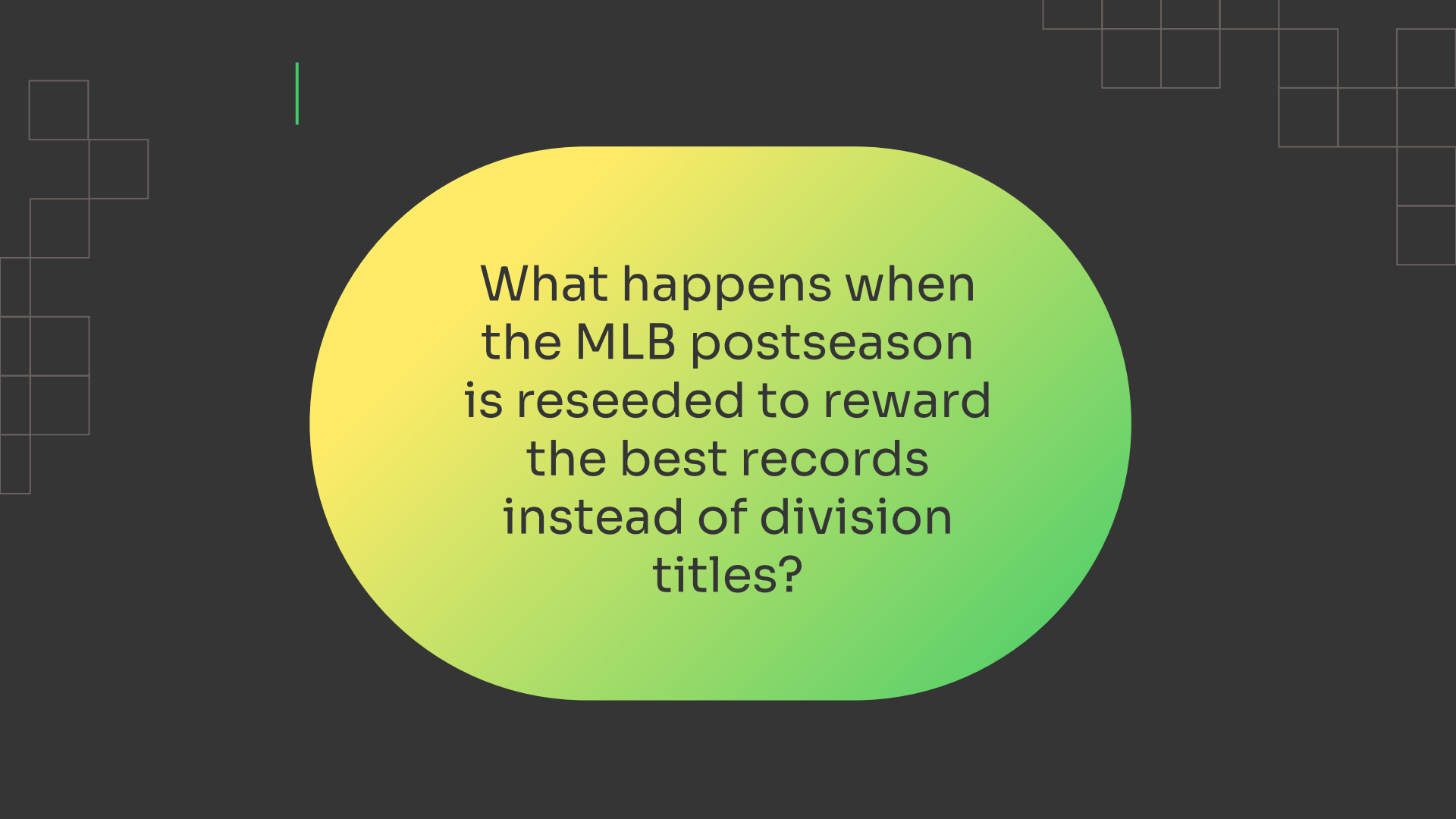
- In the MLB postseason, teams are ranked and placed into the playoff bracket based on their division titles.
- Division winners automatically receive higher rankings (seeds) than some teams with better records, giving them advantages like easier matchups and home-field.
- As a result, strong teams from tougher divisions can start the playoffs at a disadvantage.
- Wild card teams can have better records but lower seeds



Why this matters:

- The current playoff system can reward weaker divisions and penalize stronger ones.
- A 162-game season gives enough data to accurately measure team performance
- Using a Bradley–Terry statistical model helps capture each team's true performance rather than just raw total wins
- Reseeding by record could make the postseason more balanced and data-driven.



A decorative background featuring a dark gray field with a grid of white squares. The grid is composed of several clusters of squares, some of which are partially cut off by the edges of the frame. A single vertical green line is positioned near the top left of the central oval.

What happens when
the MLB postseason
is reseeded to reward
the best records
instead of division
titles?

Creating our Data:

- Source: `mlb_schedule(season = 2025)`
from `baseballr`
- Filter: `status = "Final" or "Completed Early"`
+ `game_type = "R"`
- Drop: non-MLB teams (`league_id ≠ 103, 104`)
- Join with `mlb_teams()` for `team_id`, league (AL/NL)
- Normalize: Team names for consistency
- Output: complete home/away win-loss dataset



New Playoff Format

New AL Playoff Seeds

1. Toronto Blue Jays (1)*
2. New York Yankees (4)
3. Seattle Mariners (2)*
4. Boston Red Sox (4)
5. Cleveland Guardians (3)*
6. Detroit Tigers (6)

Number in () is original seed in 2025 MLB Playoffs.

* indicates Division Winner

New NL Playoff Seeds

1. Milwaukee Brewers (1)*
2. Philadelphia Phillies (2)*
3. Los Angeles Dodgers (3)*
4. Chicago Cubs (4)
5. San Diego Padres (5)
6. Cincinnati Reds (6)

NL Standings unchanged from Actual 2025 MLB Playoff Seeds

Playoff Matchup Restructuring:

From Wild Card (WC) Round to Division Series:

Lowest remaining seed after WC Round plays 1 seed. Other WC Round winner plays 2 seed.

Example: If the 6 seed Reds beat the Dodgers in the WC Round, they would face the Brewers in the Division Series.

- In old format, the winner of the 3v6 WC matchup always played the 2 seed

Data Modeling

- Bradley–Terry model estimates team “ability” (β_i)
- Win probability: $P(A \text{ beats } B) = \frac{\exp(\beta_A)}{\exp(\beta_A) + \exp(\beta_B)}$
- Home field advantage coefficient added as well
- Fitted using the BradleyTerry2 Package
- This model captures matchup probabilities for any two teams, no matter who is playing at home.
- Created Monte Carlo functions to be able to simulate games using Bradley–Terry Coefficients.

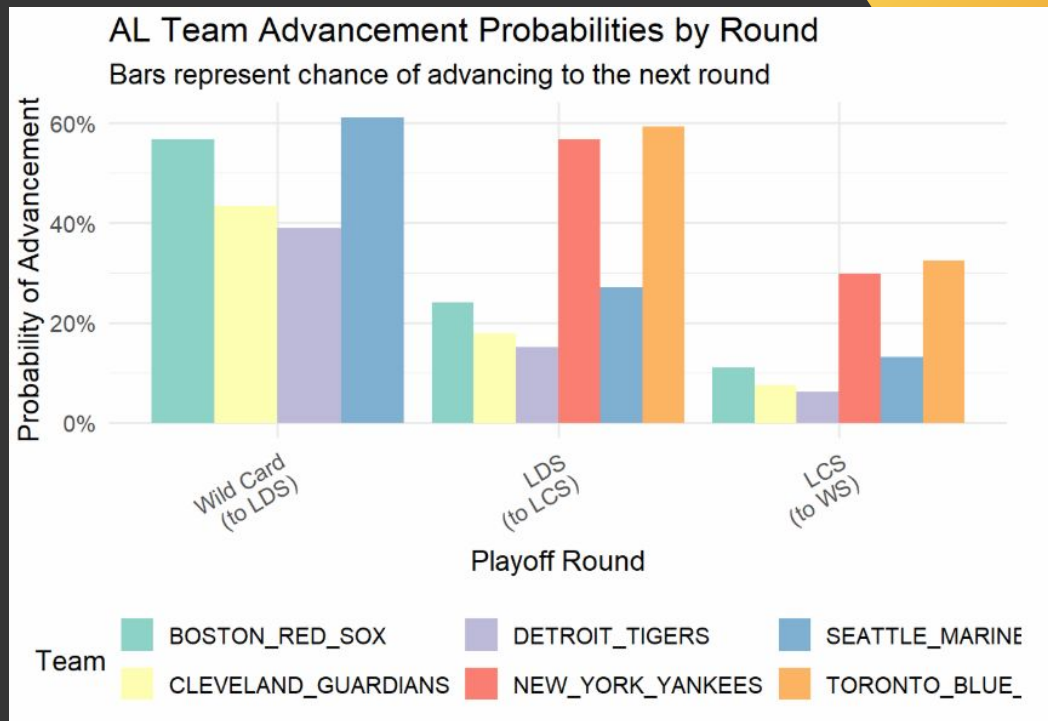
Simulation Design

- Built playoff simulations using Bradley–Terry team ability scores as input win probabilities.
- Each simulation tracked winners at every round for Wild Card, LDS, LCS, and WS
- Calculated two main outputs:
 - Matchup-specific odds: probability a team wins a given round matchup.
 - Advancement odds: probability a team advances to a given round (LDS, LCS, WS).
- Reseeded and Restructured Format
 - Reseeded format (purely by regular-season record).
 - Restructured Division Round (1 seed faces lowest remaining seed)
- Computed league-specific results for AL and NL separately.

Key Result | Visualization

AL Playoff Probabilities

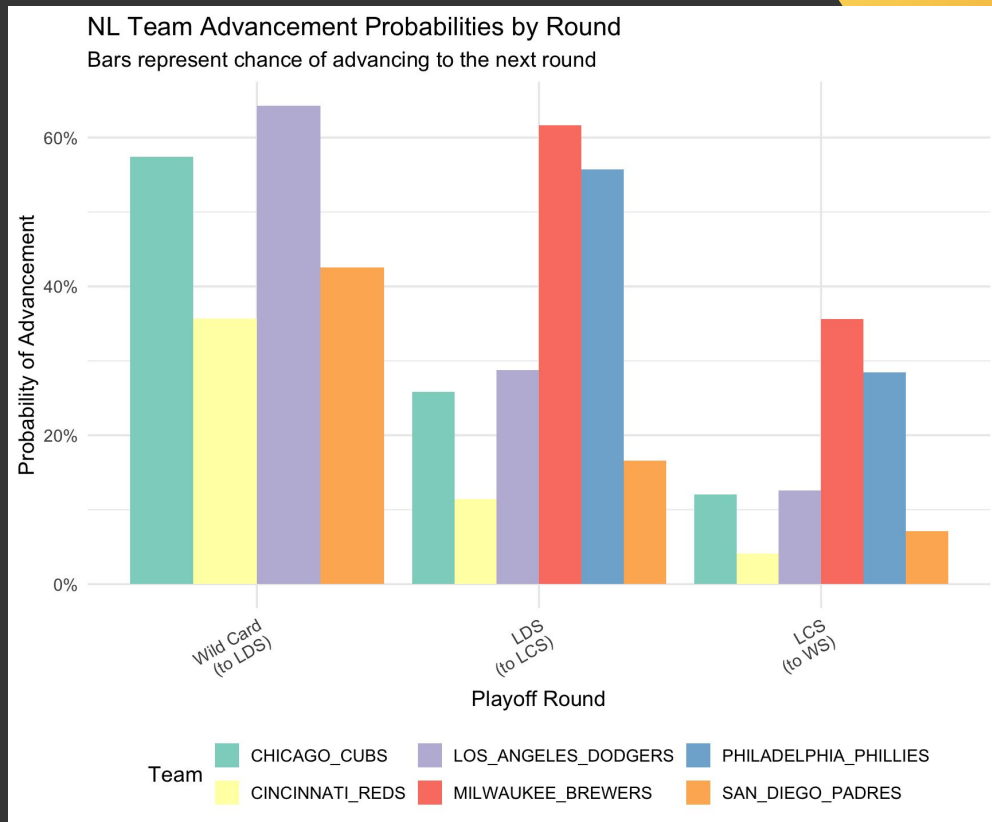
- AL simulation results show clear stratification by team strength.
- “Advance to LDS”, “Advance to LCS”, and “Advance to WS” probabilities computed via simulation.
- Top 2 seeds have higher odds to go further
 - Yankees and Blue Jays show highest advancement rates.
 - Wild Cards made it past the LDS Round much less often



Key Result | Visualization

NL Playoff Probabilities

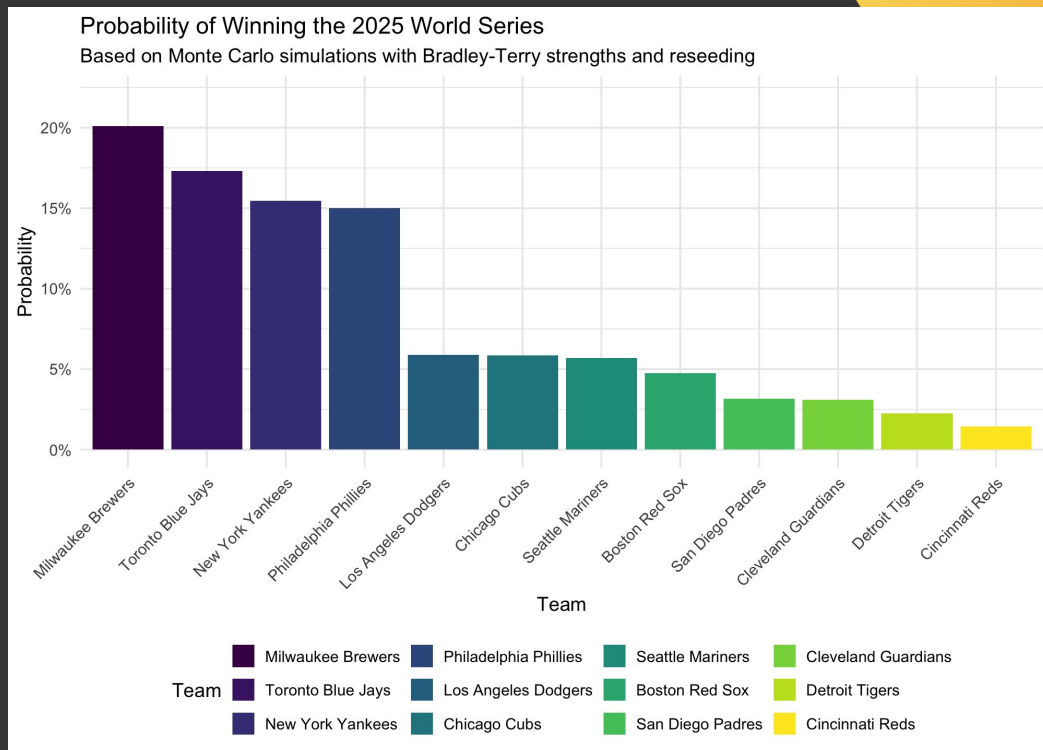
- NL results mirror AL trends higher seeds advancing more often than the Wild Card teams.
- Brewers and Phillies (Top 2 seeds) mirror AL top teams with strongest odds of advancing
- Actual MLB Champs Dodgers only made it to World Series 12.6% of simulations
- Reinforces that reseeding benefits high-performing regular season teams.



Key Result | Visualization

World Series Playoff Probabilities

- Yankees Example
- Post reseeding and restructuring brackets using 2,000 MC simulation results:
- Brewers, Blue Jays, Yankees and Phillies have highest probabilities.
 - These teams won 67.85% of titles in our simulation



Conclusion

- Main takeaway:
 - Reseeding MLB playoffs by record gives best MLB teams from regular season best chance to win the title.
 - Division winner perks shrink: weaker division winners no longer gain automatic high ranks or home-field advantages.
 - Strong wild-card teams (with better records) gain fairer matchups and higher simulated advancement probabilities.
- Statistical insight: The Bradley–Terry model effectively quantifies each team's ability and captures home advantage (~4.5% edge).
- Analytical value: Shows how probabilistic modeling can evaluate sports policy decisions beyond simple win-loss stats.
- Limitations: Assumes constant team strength throughout the year.
 - No injuries, moral boost, or changed team performance.

Next steps:

- Extend model across multiple seasons for stability.
- Compare our results to actual MLB outcomes.
- Simulate alternative playoff sizes or keep set matchups after each round.
- Change how many games each series plays.
- Alternate home game location advantage similar to hockey (so no team can have a 3 game stretch of hometeam advantage)
- Big picture: Based on our model and simulation, this playoff format best rewards the top teams over 162 games with a fair advantage to win the title.



Thank you